

OpenClaw Camera Application Tutorial

Before starting, please complete the following in the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw Ajetson-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Scheme Configuration Test]. This article assumes the basic environment is already prepared and only covers camera-related content.

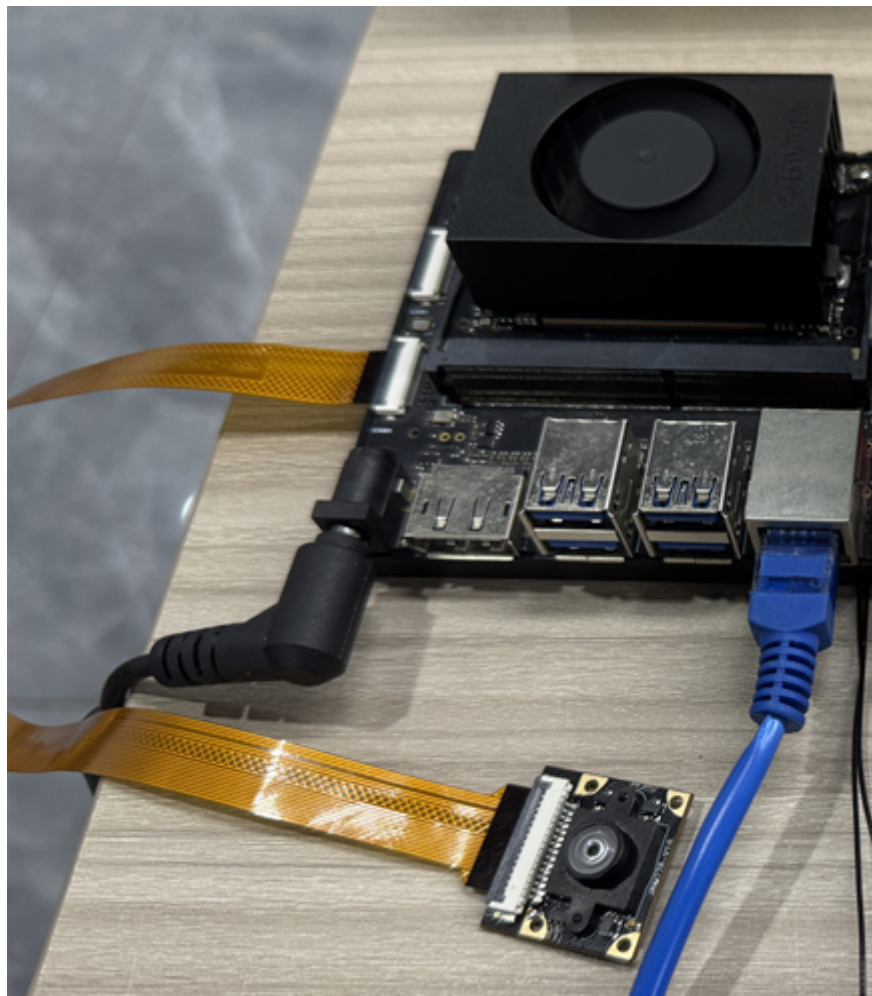
1. Tutorial Objective

This article demonstrates OpenClaw's natural language control capabilities through the CSI camera peripheral. After completing this tutorial, readers will be able to:

- Understand the basic approach of OpenClaw using the camera for photo capture and visual analysis
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice interaction
- Complete four camera cases: photo archiving, scene description, OCR recognition, object positioning and face detection
- Understand the complete chain from "user submitting a visual task" to "image capture" and "result return"

2. Hardware Preparation

- OpenClaw main control board
- CSI camera module
- Camera ribbon cable
- `[Optional] Microphone module`
- Wiring diagram



3. Software Preparation

If using WhatsApp, to upload images directly to WhatsApp, you need to enter your WhatsApp dialogue ID in the `allowFrom` field at the following path

- Open your newly created WhatsApp dialogue to see the dialogue ID



- Fill this into "allowFrom":

```
{ } openclaw.json X
.openclaw > { } openclaw.json > { } channels > { } feishu > abc chat_id
144     "commands": {
145         "native": "auto",
146         "nativeSkills": "auto",
147         "restart": true,
148         "ownerDisplay": "raw"
149     },
150     "session": {
151         "dmScope": "per-channel-peer"
152     },
153     "channels": {
154         "whatsapp": {
155             "enabled": true,
156             "dmPolicy": "allowlist",
157             "selfChatMode": true,
158             "allowFrom": [
159                 "+8615338857526"
160             ],
161             "groupPolicy": "allowlist",
162             "debounceMs": 0,
163             "mediaMaxMb": 50
164         },
165     },
166 }
```

- Enter the following command in the terminal to apply

```
openclaw gateway restart
```

4. Testing Camera Application

Before starting this tutorial, please complete a basic `openclaw tui` dialogue self-check first; if you haven't done so, see [05-Three Dialogue Scheme Configuration Test]. Also, please confirm that the visual model has been configured, refer to [02-OpenClaw Ajetson-KEY Configuration] and [03-OpenClaw Model Switching]. After passing, proceed to the camera-specific test.

Enter the following command in the terminal to enter TUI:

```
openclaw tui
```

```
OpenClaw 2026.3.12 (6472949)
Your terminal just grew claws-type something and let the bot pinch the busywork.

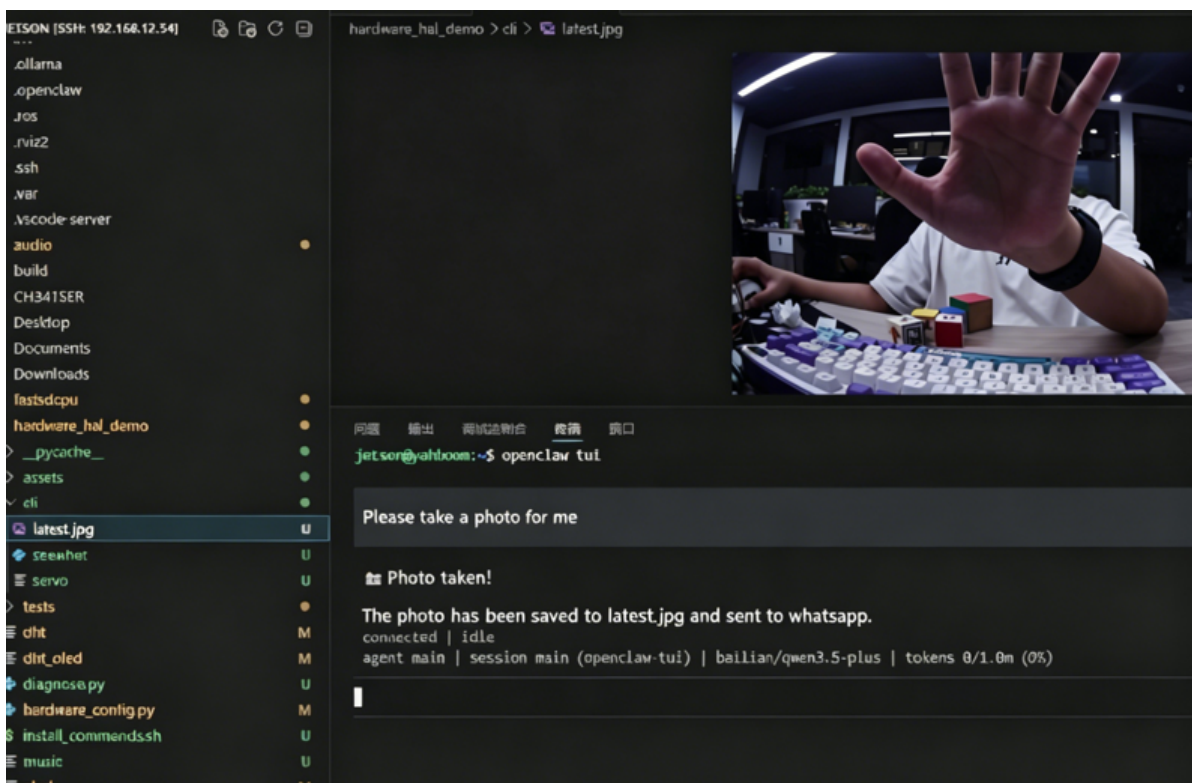
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

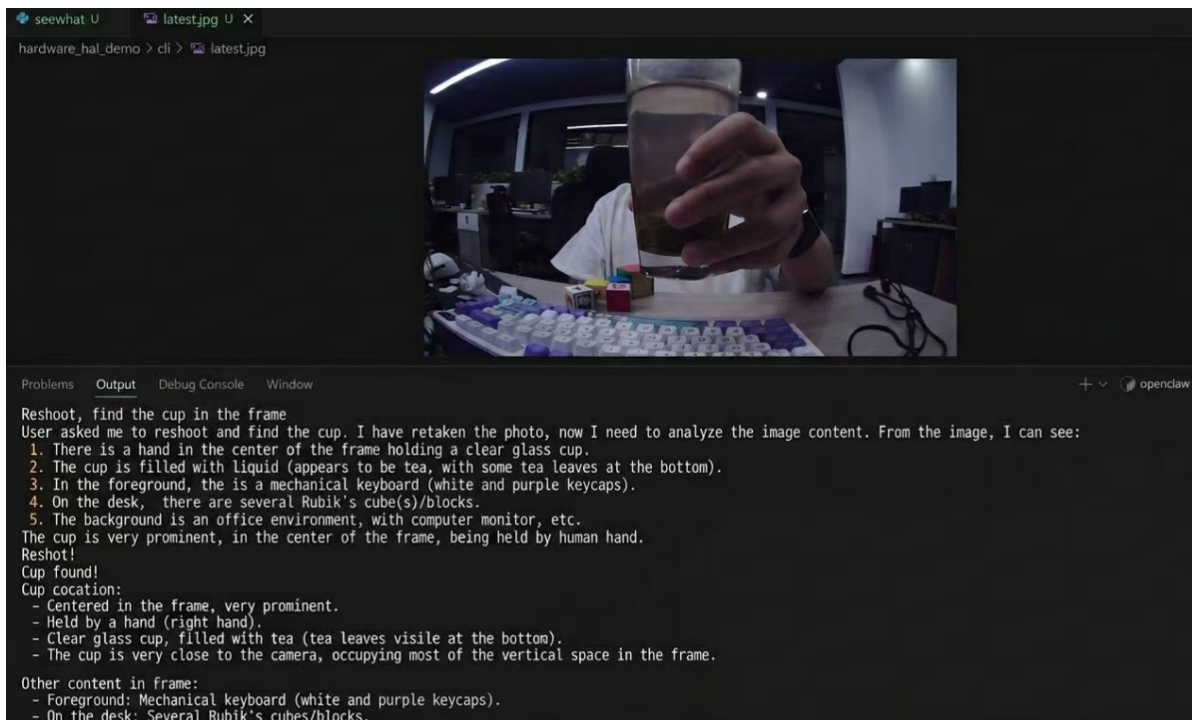
session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

You can try entering the following content in the terminal:

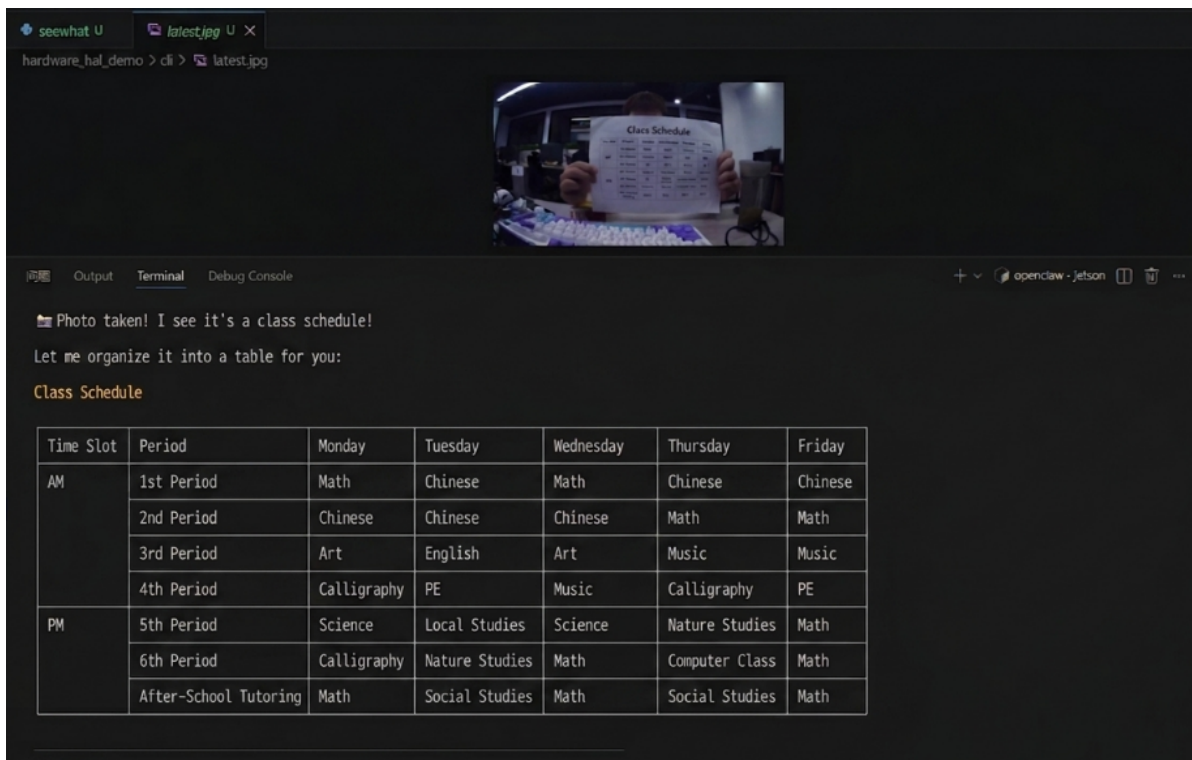
- Help me take a photo
- Recognize the text in this image



- Find the cup in the frame (because the model will reference context, you need to remind it to retake the photo, otherwise it might use the first photo)



- Retake a photo and organize the text in it into a table



5. Three Schemes for Communicating with OpenClaw

Scheme	Suitable Scenarios	Advantages	Recommended Use
WhatsApp	Remote control, classroom demonstration	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First run through the visual chain
Voice code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration scheme

If you have the optional AI voice module + speaker, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue does not need to run this repeatedly.

- Modify the TTS toggle parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter controls whether to broadcast. After modifying and saving, run the following program in the terminal
```

- Enter in terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
jetson@yahboom:~$ cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 whatsapp Scheme

For WhatsApp connection steps, please directly refer to [05-Three Dialogue Scheme Configuration Test]. This section only includes dialogue examples suitable for the camera tutorial.

After opening WhatsApp, you can usually also see the photos and recognition results triggered by `openclaw tui` earlier synchronized over.

19:55



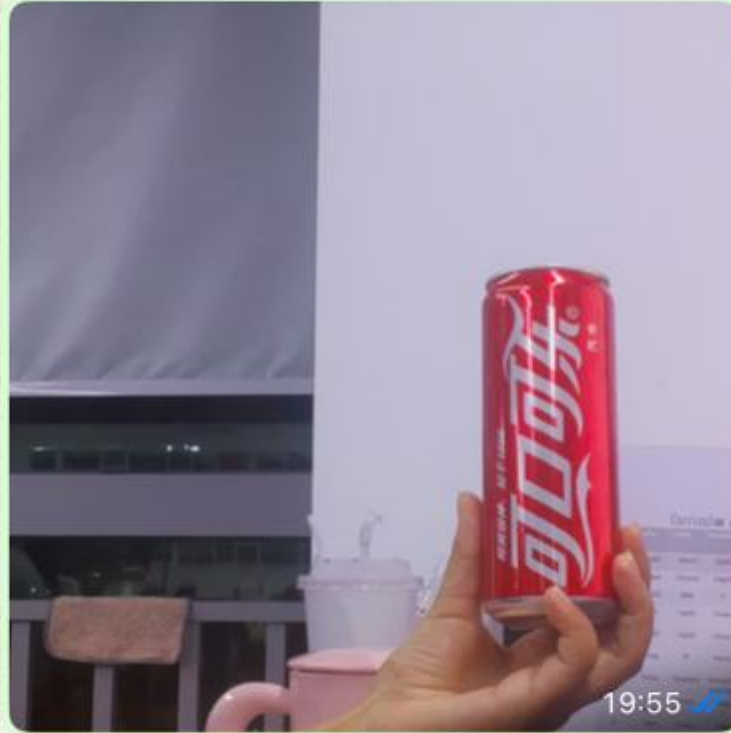
< 2



+86 153 3885 7526 (你)

给自己发消息

Look at what's in front of 19:55 ✓✓



I can see the scene in front of the camera:

- A hand is holding a **red Coca-Cola can** in the center
- To the right: a small **green desk fan** and a **water bottle**
- In the background: a **curriculum schedule** paper posted on the wall
- On the left: what looks like a **purple phone case** and a **window with blinds**
- Various items on the desk including a **pink mug** and some containers

The scene appears to be a desk or



5.2 TUI Scheme

If you have completed the TUI self-check in Part 0, simply enter the following command in the terminal to continue the conversation:

```
openc1aw tui
```

Then you can directly enter camera-related commands, for example:

- Take a photo of the current scene
- Describe what the camera sees
- Check if there are any faces

You can add your own TUI dialogue screenshots here.

5.3 Voice Code Scheme

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. After wake-up, you can issue voice commands directly, for example:

- Help me take a photo
- Look at what's ahead
- Help me read the text in the frame

You can add your own voice wake-up and voice control screenshots here.

6. Comprehensive Cases

Note: The following are all demonstrated using the `openc1aw tui` method. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: Photo Archiving

Experimental Objective

Have OpenClaw first complete the most basic photo-taking function and save the image.

Effect Description

After execution, a photo will be generated locally, typically saving both the original photo file and a `latest.jpg` alias for the most recent photo. This case is suitable for verifying the image capture chain.

Example Instructions

- Help me take a photo
- Capture the current scene
- Take a photo now for records

6.2 Case 2: Scene Description

Experimental Objective

Have OpenClaw provide a brief description of the current scene after taking a photo.

Effect Description

OpenClaw can first complete the photo capture, then call the visual analysis script to provide a one-sentence description of the image content, such as identifying desktops, people, food, or common objects.

Example Instructions

- Look at what's ahead
- Help me describe the current scene
- After taking a photo, tell me what's mainly in the frame

6.3 Case 3: OCR Table Recognition

Experimental Objective

Have OpenClaw recognize text in images and organize it into a table.

Effect Description

This case is suitable for photographing labels, cards, packaging, or simple documents, then extracting the text. It can output plain text or continue to organize it into a table.

Example Instructions

- Recognize the text in this image
- After taking a photo, organize the text into a table
- Find the cup in the frame
- Check if there are any faces
- Help me locate the phone on the right side of the desk